// **Exercise 2: Implementing the Factory Method Pattern**

```
interface Document {
   void open();
}
class WordDocument implements Document {
   public void open() {
      System.out.println("Opening Word Document");
   }
}
class PdfDocument implements Document {
   public void open() {
      System.out.println("Opening PDF Document");
   }
}
class ExcelDocument implements Document {
   public void open() {
      System.out.println("Opening Excel Document");
   }
}
abstract class DocumentFactory {
   public abstract Document createDocument();
}
class WordFactory extends DocumentFactory {
   public Document createDocument() {
      return new WordDocument();
   }
}
class PdfFactory extends DocumentFactory {
   public Document createDocument() {
      return new PdfDocument();
   }
}
class ExcelFactory extends DocumentFactory {
   public Document createDocument() {
      return new ExcelDocument();
   }
}
public class FactoryMethod {
   public static void main(String[] args) {
```

```
        DocumentFactory wordFactory = new WordFactory();
        Document wordDoc = wordFactory.createDocument();
        wordDoc.open();
        DocumentFactory pdfFactory = new PdfFactory();
        Document pdfDoc = pdfFactory.createDocument();
        pdfDoc.open();
        DocumentFactory excelFactory = new ExcelFactory();
        Document excelDoc = excelFactory.createDocument();
        excelDoc.open();
    }
}
```

Output: